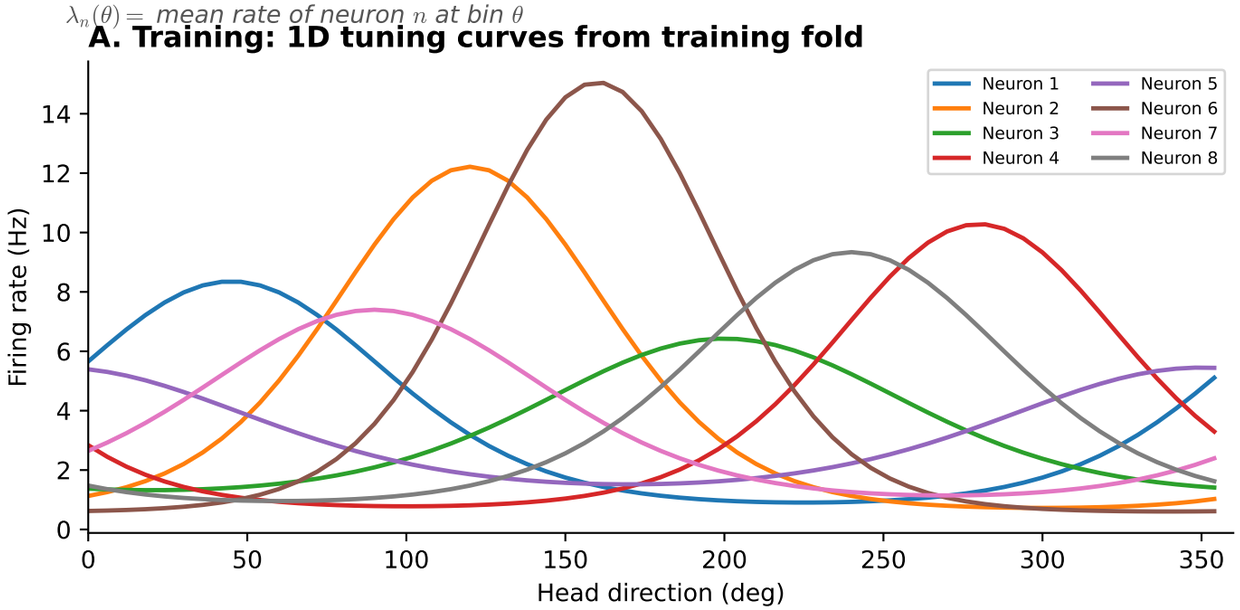
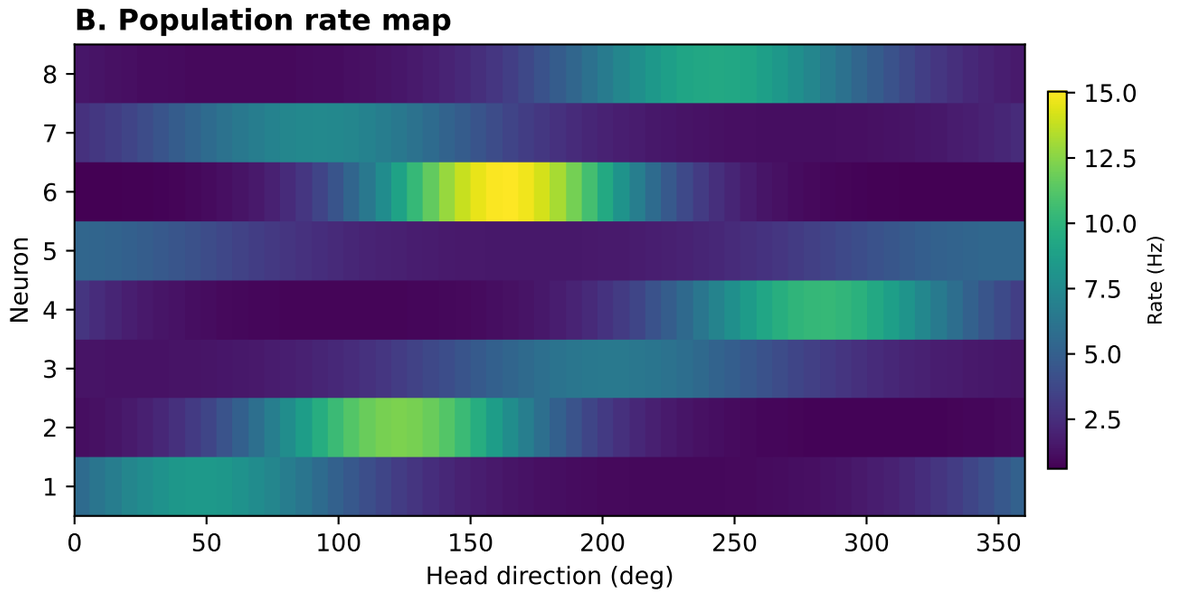
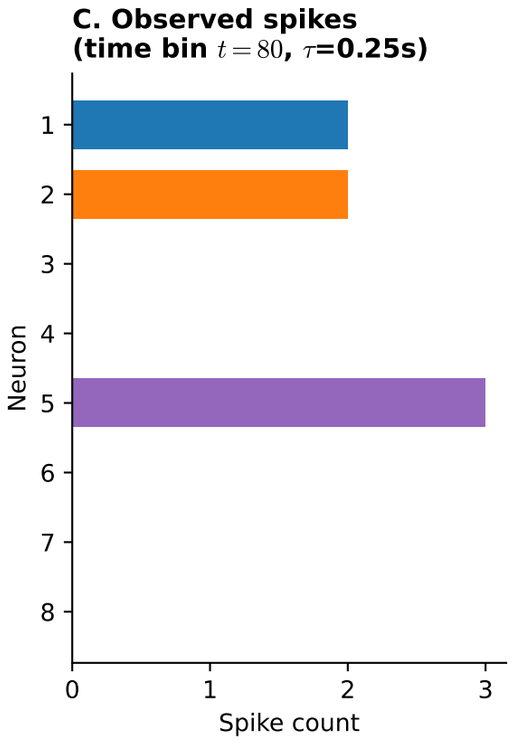
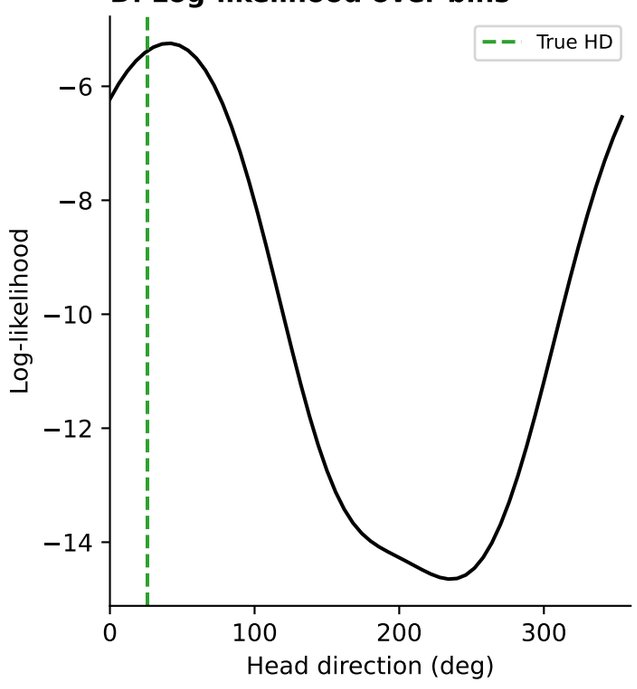
1D Bayesian Population Decoding

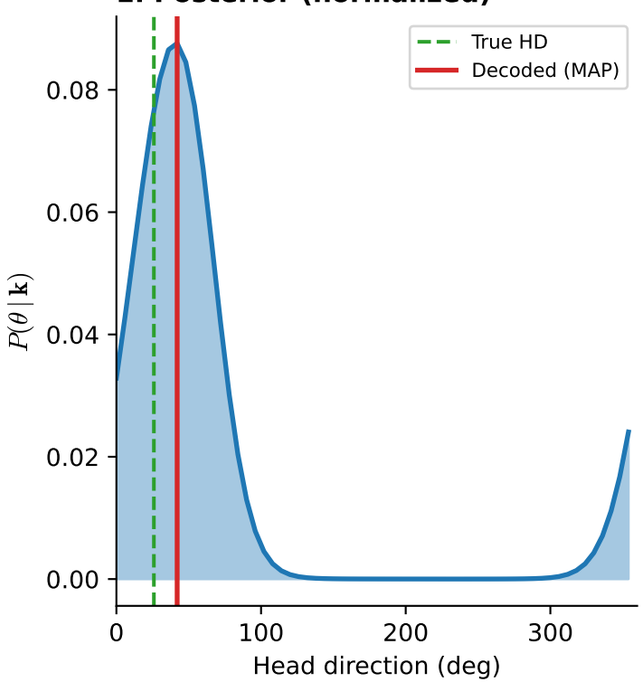
$\lambda_n(\theta) =$ *mean rate of neuron* n *at bin* θ

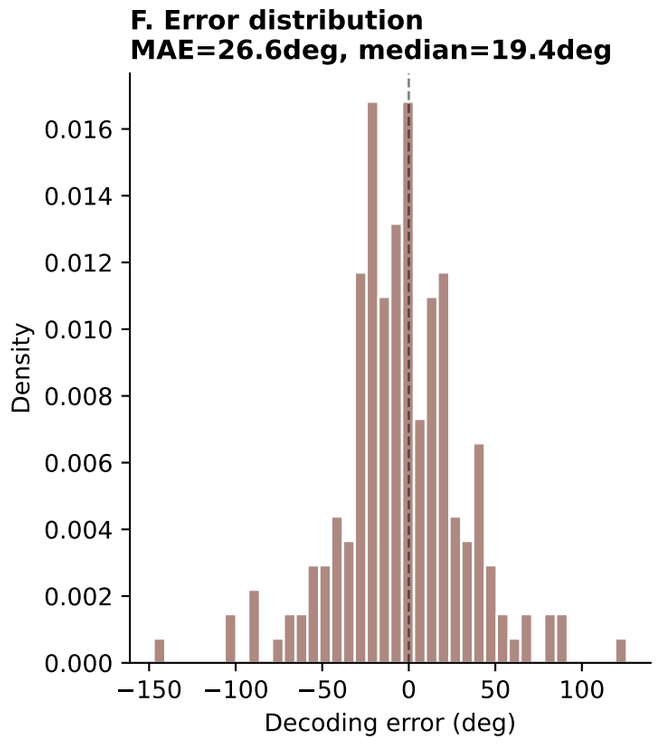


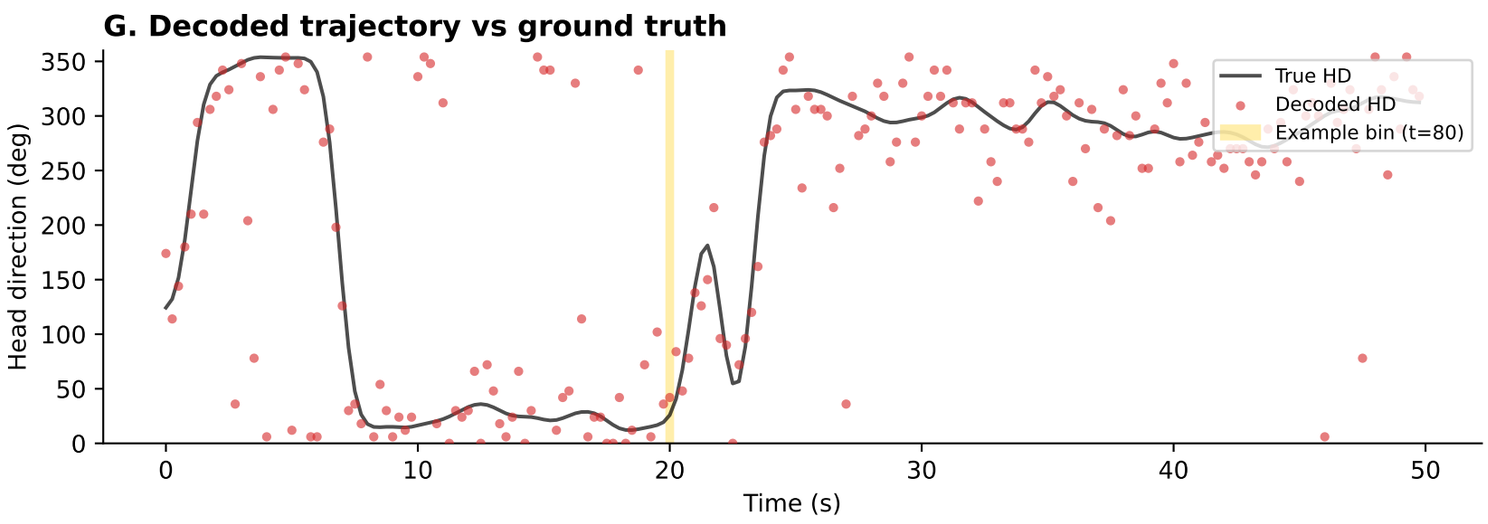












Poisson likelihood :

$$\log P(\mathbf{k} \mid \theta) = \sum_{n=1}^{N} [k_n \log(\lambda_n(\theta)\,\tau) - \lambda_n(\theta)\,\tau]$$

Posterior :

$$P(\theta \mid \mathbf{k}) = \frac{P(\mathbf{k} \mid \theta)\,P(\theta)}{\sum_{\theta'} P(\mathbf{k} \mid \theta')\,P(\theta')}$$

Decoded stimulus :

$$\hat{\theta} = \arg\max_{\theta} P(\theta \mid \mathbf{k})$$

where k_n = spike count of neuron n,
$\lambda_n(\theta)$ = tuning curve rate,
τ = time bin width, $P(\theta)$ = flat prior